

Keshav Rajput

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Data engineer / data scientist, new grad. MS Computer Science, NYU Tandon, May 2026. PySpark, SQL, pandas, R.
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EDUCATION

New York University, Tandon School of Engineering
Master of Science (MS) in Computer Science

Sep 2024 - May 2026
GPA 3.67/4.0

Indraprastha Institute of Information Technology (IIIT) Delhi

Bachelor of Technology (B.Tech) in Computer Science and Applied Mathematics, minor in Economics

2020 - 2024
CGPA 8.19/10

Coursework: Big Data, Principles of Database Systems, Information Visualization, Foundations of Data Science, Statistical Inference, Machine Learning

PROJECTS

NYC AutoDDG | Python, PySpark, pandas, PyArrow, Ollama | github.com/happyc0der/nyc-autoddg

2026

- Single-pass PySpark profiler (types, statistics, semantic types, data-quality flags) for 100 NYC Open Data datasets: 4.8M rows in 2.3 min, down from 12; checked against a pandas oracle with pytest.
- Benchmarked Spark vs pandas on an M4 Pro and Databricks serverless over Parquet inputs: crossover at 1.7M rows, near-linear to 8 cores, pandas 14x faster on small files.
- A local qwen3:14b wrote AutoDDG-style descriptions from the profile; 97% of checkable claims matched, and search Recall@1 rose from 0.605 to 0.728.

TrafficFlowBench (in progress) | Python, LightGBM | github.com/happyc0der/trafficflowbench

2026

- Kaggle TrafficFlowBench (IEEE Big Data Cup 2026): public leaderboard 0.845 vs the 0.553 baseline, up from 0.67365 on my first submission; final ranking pending.
- Four-task data pipeline: same-link interpolation for missing cells, LightGBM time-series queue forecasters trained out of core on 56M rows with causal features, an exact NNLS origin-destination solver.
- Found a window-selection regularity that lifted queue-onset IoU from 0.57 to 0.93 and reported it on the forum; reproduce.sh rebuilds the submission bit-for-bit in about two hours.

Kaggle S6E9, EV purchases | Python, LightGBM

2026

github.com/happyc0der/kaggle-s6e9-ev-purchases

- Kaggle Playground S6E9 (ROC AUC): out-of-fold 0.94648, public leaderboard 0.94649 against a leader at 0.94674; final ranking pending.
- Feature engineering with nested target encodings over the synthetic generator's artifacts on 668,665 rows, 20-fold cross-validation; logit blend of LightGBM, CatBoost and a 15-seed MLP.
- 192 experiments judged by paired DeLong hypothesis tests; label resampling showed a perfect model scores 0.9459 +/- 0.0003, so the gap left is label noise.

Biohub cell tracking (in progress) | Python, PyTorch, NumPy, SciPy, polars

2026

Builds on Igor Zharov's public notebook (Apache-2.0); the harness, classifiers and experiments are mine.

- Kaggle Biohub Cell Tracking: public leaderboard 0.952 against 0.947 for the public notebook it builds on, on free Kaggle GPU; final ranking pending.
- Offline harness runs GPU inference once, then re-tunes post-processing locally against the official scorer on 40 held-out videos, matching the notebook to +0.00011.
- 3D CNN trained on 134 annotated divisions prunes false forks: predicted +0.0047 on a pre-registered holdout, measured +0.005 on the leaderboard twice.

COMPETITIONS

- **AI x GTM Hackathon (Block Convey), Revenue Intelligence track**, 2026. Entry is Ledger, a pipeline-review agent that checks every claim against the CRM. Event on 2026-10-17.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Parquet, time-series forecasting, econometrics (fixest), Streamlit, Altair, matplotlib

ML: PyTorch, deep learning, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, NumPy, pandas, polars

Tools: Git, CI/CD (GitHub Actions), Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale, Claude Code

LEADERSHIP AND ACTIVITIES

- Open source: ephys-mcp on PyPI and the MCP Registry (24 tools, 52 tests); six fixes with regression tests to hawk-py
- Events head, Finnextia (finance club), IIIT Delhi; organised "The Critical Bid"